

NASP Neuron Core Operational Amplifier Requirements

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Revision History

Revision	Date	Editor	Details
0.1	09.03.2026	Valeriy Shunkov	Initial draft
0.2	20.03.2026	Valeriy Shunkov	Parametrics put into a table

Scope

This specification defines a complete set of traceable requirements for the NASP Neuron Core Operational Amplifier (Operational Amplifier).

Description

NASP Neuron Core Operational Amplifier (Operational Amplifier) is the operational amplifier designed to be the core of a neuron in an analog neural network (ANN) of the NASP Silicon Platform. The Operational Amplifier is part of the platform, which also includes support and test circuitry for ANNs, as well as interface circuits for signal generation and acquisition.

Detailed interface definitions, parameter definitions, measurement methods, and operating-condition details are specified in the Operational Amplifier block specification. Detailed requirement-to-verification mapping and verification procedures are specified in the Operational Amplifier verification plan.

Document Structure

This document is organized into the following major sections:

Scope defines the purpose of the document, introduces the Operational Amplifier, and explains how this requirements file relates to the block specification and verification plan.

System-Level defines environmental, absolute-maximum, and process-level assumptions that apply to the Operational Amplifier.

Product-Level defines the intended role of the Operational Amplifier in the product, its external interfaces, and the main internal building blocks expected to be present.

Design Flow defines the required design deliverables, implementation constraints, and layout/schematic conventions to be followed during development.

Functional Requirements defines the behavior of the Operational Amplifier, including operating modes, control-state constraints, auxiliary-switch behavior, and startup/reset behavior.

Parametric Requirements define the electrical performance targets of the Operational Amplifier, including input/output operating ranges, gain and stability, offset, and current consumption.

Robustness and Reliability defines requirements related to fault tolerance, overload tolerance, latchup, and compatibility with top-level protection targets.

Testability defines the features required for observing, controlling, and resetting the Operational Amplifier during validation and production test.

Appendix A. Glossary defines abbreviations and project-specific terms used throughout the document.

References

NASP Silicon Platform Product Requirements Document.

NASP Silicon Platform Integrated Circuit Specification Document.

Operational Amplifier Block Specification.

Operational Amplifier Verification Plan.

Naming Conventions

Requirements and Preconditions Naming

Preconditions are numbered as "**P-OPA-#**", where # is its decimal number. Preconditions have no type.

Requirements are numbered as "**R-OPA-type-#**", where "type" is the type of a requirement, and # is its decimal number.

Types of requirements are:

SYS — system-level.

PRD — product-level.

PTF — process technology and design flow.

FNC — functional.

PRM — parametric.

REL — robustness and reliability.

TST — testability

Verification Type Naming

Requirements are tagged with one or more verification tags from the following set:

VbA — verification by analysis. Used when compliance is demonstrated by calculation, derivation, or analytical evaluation.

VbR — verification by review. Used when compliance is demonstrated by inspection of documentation, schematics, layouts, models, or reports.

VbS — verification by simulation. Used when compliance is demonstrated by circuit or behavioral simulation.

VbT — verification by test. Used when compliance is demonstrated by direct measurement on silicon.

VbI—verification by indirect test. Used when compliance is demonstrated by an indirect observable correlated with the required behavior or parameter.

System-Level

Absolute Maximum Ratings

P-OPA-001: Operational Amplifier pins shall be kept between -0.3 and 1.4V at all times.

Environmental

R-OPA-SYS-001 [VbS, VbT]: Operational Amplifier shall be operational and compliant with all requirements in the ambient temperature range from -40C to 125C.

R-OPA-SYS-002 [VbA, VbS]: Device reliability and design margins shall support junction temperatures up to 150C.

Process Definition

P-OPA-002: Operational Amplifier shall be designed at the GlobalFoundries 55 nm LPX process.

P-OPA-003: Operational Amplifier shall use metal stack 6_00_02_00_LB

Product-Level

P-OPA-004: Operational Amplifier shall be accompanied by a set of bias blocks suitable for analog neural network generation for a network with a maximum of 4096 Operational Amplifiers.

R-OPA-PRD-001 [VbR]: Operational Amplifier shall have a pre-shrink silicon footprint of no more than 1000 square microns.

R-OPA-PRD-002 [VbS]: Unless explicitly stated otherwise in a specific requirement, Operational Amplifier shall be operational and compliant with all requirements in the supply voltage range from 1.08V to 1.32V.

R-OPA-PRD-003 [VbS]: Unless explicitly stated otherwise in a specific requirement, Operational Amplifier shall be operational, stable, and compliant with all requirements for load capacitance up to 1pF.

Interfaces

Functional Interfaces

R-OPA-PRD-004 [VbR]: Operational Amplifier shall have a supply pin named avdd1p2.

R-OPA-PRD-005 [VbR]: Operational Amplifier shall have a ground pin named agnd.

R-OPA-PRD-006 [VbR]: Operational Amplifier shall have two analog voltage input pins named vinp and vinn.

R-OPA-PRD-007 [VbR]: Operational Amplifier shall have an analog voltage functional output pin named vout.

R-OPA-PRD-008 [VbR]: Operational Amplifier shall have a current reference input pin named in0u25_oa.

R-OPA-PRD-009 [VbR]: Operational Amplifier shall have an analog pin named vbase.

R-OPA-PRD-010 [VbR]: Operational Amplifier shall have an analog pin named vfeed.

R-OPA-PRD-011 [VbR]: Operational Amplifier shall have digital control input pins named d_en_oa, d_az_oa, and d_inf_oa.

Test Interfaces

R-OPA-PRD-012 [VbR]: Operational Amplifier shall have an analog voltage test output pin named vtest.

R-OPA-PRD-013 [VbR]: Operational Amplifier shall have a digital test reset input pin named d_treset_oa.

R-OPA-PRD-014 [VbR]: Operational Amplifier shall have a digital test clock input pin named d_tcki and a digital test clock output pin named d_tcko.

R-OPA-PRD-015 [VbR]: Operational Amplifier shall have a digital test data input pin named d_tdi and a digital test data output pin named d_tdo.

Composition

R-OPA-PRD-016 [VbR]: Operational Amplifier shall include the OPA core.

R-OPA-PRD-017 [VbR]: Operational Amplifier shall include the internal reference-generation circuitry generating all necessary internal reference currents and voltages.

R-OPA-PRD-018 [VbR]: Operational Amplifier shall include the internal offset calibration circuitry.

R-OPA-PRD-019 [VbR]: Operational Amplifier shall include the internal control circuitry.

R-OPA-PRD-020 [VbR]: Operational Amplifier shall include testing circuitry.

Design Flow

Deliverables

Block Deliverables

R-OPA-PTF-001 [VbR]: Symbol view of the Operational Amplifier shall be delivered.

R-OPA-PTF-002 [VbR]: Schematic view of the Operational Amplifier shall be delivered.

R-OPA-PTF-003 [VbR]: Functional view of the Operational Amplifier shall be delivered.

R-OPA-PTF-004 [VbR, VbS]: Verilog functional behavioral model of the Operational Amplifier shall be delivered for use in digital verification.

R-OPA-PTF-005 [VbR, VbS]: Verilog-A parametric behavioral model of the Operational Amplifier shall be delivered for use in analog verification.

R-OPA-PTF-006 [VbR, VbA]: Mathematical parametric behavioral model of the Operational Amplifier shall be delivered for use in the NASP Compiler.

R-OPA-PTF-007 [VbR]: Layout view of the Operational Amplifier shall be delivered.

R-OPA-PTF-008 [VbR, VbA]: Layout view of the Operational Amplifier shall be DRC, ERC, and LVS clean.

R-OPA-PTF-009 [VbR]: Abstract view of the Operational Amplifier shall be delivered.

R-OPA-PTF-010 [VbR]: RC-extracted view of the Operational Amplifier shall be delivered. RC extraction shall be performed for typical process, voltage, and temperature conditions.

Supporting Deliverables

R-OPA-PTF-011 [VbR]: A set of testbenches shall be delivered.

R-OPA-PTF-012 [VbR]: Each testbench shall include at least maestro and schematic views.

R-OPA-PTF-013 [VbR]: The Operational Amplifier specification shall be delivered as an AsciiDoc file and a PDF file.

R-OPA-PTF-014 [VbR]: The Operational Amplifier verification plan shall be delivered as an AsciiDoc file and a PDF file.

R-OPA-PTF-015 [VbR]: A user guide shall be delivered as an AsciiDoc file and a PDF file.

Schematic Design

R-OPA-PTF-016 [VbR]: Only LVT, normal VT, and HVT PMOS core transistors shall be used.

R-OPA-PTF-017 [VbR]: Only normal VT and HVT NMOS core transistors shall be used.

R-OPA-PTF-018 [VbR]: Only opppc_2k resistors shall be used.

R-OPA-PTF-019 [VbR]: Only apmom capacitors shall be used.

R-OPA-PTF-020 [VbR]: Operational Amplifier shall use DFT-Automotive option for basic-elements

symbols.

R-OPA-PTF-021 [VbR]: Transistor names shall comply with the following convention: [A][T][Q][number], where [A] is "L" for LVT transistors or "H" for HVT transistors, [T] is "N" for NMOS or "P" for PMOS, [Q] is "C" for cascodes, "S" for switches, or "D" for dummies or active diodes.

R-OPA-PTF-022 [VbR]: Nominal static operating currents for each connection to the supply and ground shall be annotated in the schematic.

R-OPA-PTF-023 [VbR]: Each analog net vulnerable to digital switching noise shall be identified and marked in the schematic.

R-OPA-PTF-024 [VbR]: Each major functional subcircuit of the Operational Amplifier shall be identified and labeled in the schematic.

R-OPA-PTF-025 [VbR]: Each pin of the Operational Amplifier shall be annotated in the schematic with its functional purpose and nominal signal type.

R-OPA-PTF-026 [VbR]: Critical device groups requiring symmetry, common-centroid placement, or close matching in layout shall be annotated in the schematic with the intended layout constraint.

R-OPA-PTF-027 [VbR]: Each path between the supply and ground rails shall include at least one high-Vt transistor.

Layout Design

R-OPA-PTF-028 [VbR]: Operational Amplifier shall use DFT-Automotive option for basic-elements pcells.

R-OPA-PTF-029 [VbR]: Operational Amplifier shall have 100% multi-via coverage.

R-OPA-PTF-030 [VbR]: Operational Amplifier shall use metal layers M1-M6.

R-OPA-PTF-031 [VbR]: Operational Amplifier shall use metal layers M1-M2 for supply and ground rails.

R-OPA-PTF-032 [VbR]: Operational Amplifier signal pins shall follow the project-defined digital routing conventions.

R-OPA-PTF-033 [VbR]: Operational Amplifier signal pins shall be located at the vertical edges of the layout.

R-OPA-PTF-034 [VbR]: Operational Amplifier shall use different metal layers for analog and digital pins.

R-OPA-PTF-035 [VbR]: Each analog net vulnerable to digital switching noise shall be shielded in the layout.

R-OPA-PTF-036 [VbR]: Operational Amplifier layout shall be drawn as a rectangle with height of 28μm.

R-OPA-PTF-037 [VbR]: Operational Amplifier layout shall allow incorporation into the standard cell field with the row height of 2.8 μ m.

R-OPA-PTF-038 [VbR]: Operational Amplifier layout shall allow abutment to other Operational Amplifiers and Bias cells.

Functional Requirements

Operating Modes

Control-State Constraints

P-OPA-005: Control inputs d_en_oa, d_az_oa, and d_inf_oa shall be generated by the control-generation circuitry of the upstream Last-stage bias block (biasc).

P-OPA-006: Prevention of forbidden or harmful combinations of d_en_oa, d_az_oa, and d_inf_oa shall be the responsibility of Last-stage bias block (biasc).

P-OPA-007: Last-stage bias block (biasc) shall ensure that the forbidden control state d_en_oa = 1, d_az_oa = 1, and d_inf_oa = 1 is never applied to the Operational Amplifier.

P-OPA-008: Last-stage bias block (biasc) shall ensure that the control state d_en_oa = 1, d_az_oa = 0, and d_inf_oa = 0 is applied only as the transient latching state during the allowed non-overlap interval between calibration and inference.

Disabled Mode

R-OPA-FNC-001 [VbR, VbS]: Operational Amplifier shall have the disabled operating mode, where all analog circuitry is disabled.

R-OPA-FNC-002 [VbR, VbS]: The disabled mode shall correspond to d_en_oa = 0. The values of d_az_oa and d_inf_oa shall be ignored when d_en_oa = 0.

P-OPA-009: Compliance with parametric requirements is not required in the disabled mode, unless explicitly stated in a requirement text.

R-OPA-FNC-003 [VbR, VbS]: Operational Amplifier shall have no uncontrolled DC paths between avdd1p2 and agnd in the disabled mode.

R-OPA-FNC-004 [VbR, VbS]: All current sources and current mirrors shall be pulled to their respective supply rails in the disabled mode.

R-OPA-FNC-005 [VbR, VbS]: The functional analog output vout shall be kept in the high-impedance state in the disabled mode.

R-OPA-FNC-006 [VbR, VbS]: The current reference input in0u25_oa shall present high impedance to the external source in the disabled mode.

Calibration Mode

R-OPA-FNC-007 [VbR, VbS]: Operational Amplifier shall have a calibration operating mode for measuring and compensating offset.

R-OPA-FNC-008 [VbR, VbS]: The calibration mode shall correspond to $d_{en_oa} = 1$, $d_{inf_oa} = 0$, and $d_{az_oa} = 1$.

P-OPA-010: Compliance with parametric requirements is not required in the calibration mode, unless explicitly stated in a requirement text.

R-OPA-FNC-009 [VbS]: The functional analog output v_{out} shall be kept in the high-impedance state in the calibration mode.

R-OPA-FNC-010 [VbR]: The offset calibration process shall be fully internal to the Operational Amplifier.

R-OPA-FNC-011 [VbS, VbI]: The offset calibration process shall require no more than 20 μs to achieve the required parametric values.

Latching Mode

R-OPA-FNC-012 [VbR, VbS]: Operational Amplifier shall have a latching operating mode intended to occur only during the transition from the calibration mode to the inference mode.

R-OPA-FNC-013 [VbR, VbS]: The latching mode shall correspond to $d_{en_oa} = 1$, $d_{az_oa} = 0$, and $d_{inf_oa} = 0$.

P-OPA-011: Compliance with parametric requirements is not required in the latching mode, unless explicitly stated in a requirement text.

R-OPA-FNC-014 [VbS]: The functional analog output v_{out} shall be kept in the high-impedance state in the latching mode.

R-OPA-FNC-015 [VbR, VbS]: In the latching mode, Operational Amplifier shall behave as in the calibration mode, with the offset calibration value latched.

Inference Mode

R-OPA-FNC-016 [VbR, VbS]: Operational Amplifier shall have the inference operating mode, in which it is fully functional.

R-OPA-FNC-017 [VbR, VbS]: The inference mode shall correspond to $d_{en_oa} = 1$, $d_{inf_oa} = 1$, and $d_{az_oa} = 0$.

R-OPA-FNC-018 [VbS, VbI]: After entering inference mode, Operational Amplifier shall remain compliant with all functional and parametric requirements for at least 200 μs .

Auxiliary Switches

R-OPA-FNC-019 [VbR, VbS]: Operational Amplifier shall include one switch between agnd and vbase, and one switch between avdd1p2 and vfeed.

R-OPA-FNC-020 [VbR, VbS]: Both auxiliary switches shall be conducting in the inference mode.

R-OPA-FNC-021 [VbR, VbS]: Both auxiliary switches shall be non-conducting in the disabled, calibration, and latching modes.

Startup and Reset

R-OPA-FNC-022 [VbS, VbI]: Operational Amplifier shall start in the disabled mode.

R-OPA-FNC-023 [VbS, VbT]: All flip-flops in the Operational Amplifier test-control circuitry shall be resettable by d_treset_oa.

Parametric Requirements

P-OPA-012: Unless explicitly stated otherwise, all parametric requirements apply in inference mode, across the specified supply-voltage range, across the specified ambient-temperature range, in all corners, and for load capacitance up to 1pF.

Requirement Name	Ta g	Parameter	Ab br	Uni t	Min Value	Nom Value	Max Value	Note
R-OPA-PRM-001	Vb S	Reference current at pin in0u25_oa	iref	nA	200	250	300	Accepted from an external sinking driver and mirrored internally
R-OPA-PRM-002	Vb S	Input common-mode voltage range	vin	mV		-10mV	(50% of the supply + 10mV)	Applies to vinp and vinn
R-OPA-PRM-003	Vb S, Vb I	Output compliant voltage range	vou t	mV	50mV		(supply - 50mV)	Must comply with all other applicable requirements in this voltage range
R-OPA-PRM-004	Vb S, Vb I	Output voltage range in the open-loop configuration	vou tol	mV	5mV		(supply - 5mV)	May be non-compliant with the open-loop gain requirement

Requirement Name	Ta g	Parameter	Ab br	Uni t	Min Value	Nom Value	Max Value	Note
R-OPA-PRM-005	Vb S, Vb I	Output source/sink current in inference mode	iout	μA	-25μA		25μA	Must comply with all other applicable requirements in this current range
R-OPA-PRM-006	Vb S	Open-loop gain	olg	dB	80			
R-OPA-PRM-007	Vb S	Gain-bandwidth product	gb w	MHz	1			
R-OPA-PRM-008	Vb S	Phase margin in the unity-gain configuration	pm	degrees	30			
R-OPA-PRM-009	Vb S	Gain margin in the unity-gain configuration	gm	dB	5			
R-OPA-PRM-010	Vb S	PSRR in the unity-gain configuration	psr r	dB	20			
R-OPA-PRM-011	Vb S	Input-referred offset voltage	voff no m	μV	-250μV		250μV	Covers all Monte Carlo variations
R-OPA-PRM-012	Vb S	1-sigma input-referred random offset voltage	voff sig ma	μV			60μV	
R-OPA-PRM-013	Vb S, Vb I	Inference-mode quiescent current	Iqn om	μA			10μA	

Requirement Name	Ta g	Parameter	Ab br	Uni t	Min Value	Nom Value	Max Value	Note
R-OPA-PRM-014	Vb S, Vb I	Calibration -mode quiescent current	Iqaz	μA			10μA	Applies only in the calibration mode
R-OPA-PRM-015	Vb S, Vb I	Disabled-mode leakage current	ileak	nA			15nA	Applies only in the disabled mode
R-OPA-PRM-016	Vb S, Vb I	Auxiliary switch closed-state resistance	rds w	kOhm			2kOhm	Applies to the switches between agnd and vbase and between avdd1p2 and vfeed.
R-OPA-PRM-017	Vb S, Vb I	Auxiliary switch open-state leakage current	ileaksw	nA			10nA	Applies to the switches between agnd and vbase and between avdd1p2 and vfeed.
R-OPA-PRM-018	Vb S, Vb I	Output-to-test switch closed-state resistance	rds wtm	kOhm			2kOhm	Applies to the switch connecting vout to vtest.
R-OPA-PRM-019	Vb S, Vb I	Output-to-test switch open-state leakage current	ileakswtm	nA			10nA	Applies to the switch connecting vout to vtest.

Robustness and Reliability

Protection

R-OPA-REL-001 [VbA, VbR]: Under conditions outside the recommended operating range but within the specified Absolute Maximum Ratings, Operational Amplifier shall not exhibit destructive failures.

R-OPA-REL-002 [VbA]: Operational Amplifier shall not exhibit destructive latchup.

R-OPA-REL-003 [VbR, VbS]: Operational Amplifier shall not exhibit destructive failure under output overload conditions causing sinking or sourcing current up to 200μA. Compliance with parametric requirements is not required.

Electrostatic Discharge

P-OPA-013: On-chip power sources feeding Operational Amplifier shall have secondary ESD protection.

R-OPA-REL-004 [VbA, VbR]: Operational Amplifier shall be compatible with achieving 2000V HBM at the top level.

R-OPA-REL-005 [VbA, VbR]: Operational Amplifier shall be compatible with achieving 500V CDM at the top level.

Testability

R-OPA-TST-001 [VbS]: The state of the internal test switch connecting vtest to vout shall be controlled through a daisy chain.

R-OPA-TST-002 [VbS, VbT]: Operational Amplifier shall have test-control circuitry supporting daisy chain connection through pins d_tcki, d_tcko, d_tdi, and d_tdo.

Appendix A. Glossary

Table 1. Operational Amplifier Terms Table

Term Abbreviation	Term Full Name	Note
CDM	Charged Device Model	Used in a context of ESD
ESD	Electrostatic Discharge	
HBM	Human Body Model	Used in a context of ESD
PSRR	Power Supply Rejection Ratio	